

Simplified Policy-Guided MPPI for Safe Manipulator Planning

Algorithm Implementation

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1 Notation

- x_t : System state at time t
- u_t : Control input at time t
- $f(\cdot)$: System dynamics function
- H : Planning horizon
- K : Number of MPPI samples
- λ : Temperature parameter for MPPI
- Σ : Noise covariance matrix
- $\ell(\cdot)$: Stage cost function
- $\ell_f(\cdot)$: Terminal cost function
- π_ϕ : Learned policy with parameters ϕ
- $\mathcal{U}$: Control constraint set
- $\mathcal{F}_{\text{safe}}$: Safety filter function
- u_t^{safe} : Safe control input applied to the robot
- `MPPIUPDATE`: Encapsulated MPPI update function

2 Algorithm Description

The Simplified Policy-Guided MPPI (PG-MPPI) algorithm combines model predictive path integral control with a learned policy to enable safe and efficient manipulator planning. This version encapsulates the MPPI sampling and update steps into a single function call for cleaner presentation.

Algorithm 1 PG-MPPI (Policy-Guided MPPI) for Safe Manipulator Planning

Require: Dynamics $x_{t+1} = f(x_t, u_t)$; horizon H ; samples K ; temperature λ ; noise covariance Σ ; stage cost $\ell(x, u)$; terminal cost $\ell_f(x)$; learned policy $\pi_\phi(a | s)$ (e.g., SAC/CD-SAC actor); control constraint set $\mathcal{U}$; safety filter $\mathcal{F}_{\text{safe}}(\cdot)$ (collision-avoidance / constraint projection).

Ensure: Control u_t^{safe} applied to the robot at each time step.

- 1: **(Offline, once)** Train policy π_ϕ in simulation (e.g., SAC/CD-SAC with goal reward + CDF-based safety shaping).
- 2: **Initialize** nominal control sequence $U = \{u_0, \dots, u_{H-1}\}$ (zeros or from π_ϕ).
- 3: **for** $t = 0, 1, 2, \dots$ **do**
- 4: Observe current state x_t .
- 5: **Policy-guided nominal (warm start)**
- 6: Predict a nominal rollout under the learned policy:
- 7: $\hat{x}_0 \leftarrow x_t$
- 8: **for** $i = 0$ to $H - 1$ **do**
- 9: $\hat{u}_i \leftarrow \mathbb{E}[\pi_\phi(\cdot | \hat{x}_i)]$ $\triangleright$ actor mean action
- 10: $\hat{x}_{i+1} \leftarrow f(\hat{x}_i, \hat{u}_i)$
- 11: **end for**
- 12: Set (or blend) the nominal sequence $U \leftarrow \{\hat{u}_0, \dots, \hat{u}_{H-1}\}$.
- 13: **MPPI update (encapsulated as a function)**
- 14: $U \leftarrow \text{MPPIUPDATE}(x_t, U; f, \ell, \ell_f, H, K, \lambda, \Sigma, \mathcal{U})$
- 15: $\triangleright$ MPPIUPDATE performs sampling rollouts, cost-to-go S_k , weight normalization, and path-integral update
- 16: **Safety filtering and execution**
- 17: $u_t^* \leftarrow u_0$
- 18: $u_t^{\text{safe}} \leftarrow \mathcal{F}_{\text{safe}}(x_t, u_t^*)$ $\triangleright$ e.g., collision-free projection/filter
- 19: Apply u_t^{safe} to the robot, obtain next state x_{t+1} .
- 20: **Receding horizon shift**
- 21: $U \leftarrow \{u_1, u_2, \dots, u_{H-1}, u_{H-1}\}$ $\triangleright$ shift left and pad
- 22: **end for**

2.1 Offline Policy Training

First, a policy π_ϕ is trained in simulation using reinforcement learning methods such as SAC or CD-SAC. The training incorporates goal-reaching rewards and CDF-based safety shaping to ensure the policy learns both task completion and safety awareness.

2.2 Online Planning Loop

During execution, the algorithm performs the following steps at each time step:

1. **Policy-guided warm start:** The learned policy generates a nominal trajectory that serves as an initial guess for the MPPI optimization, improving convergence speed.

2. **MPPI update:** The encapsulated MPPIUPDATE function performs the standard MPPI sampling, cost evaluation, weight computation, and control sequence update.
3. **Safety filtering:** The first control action from the optimized sequence is passed through a safety filter that ensures collision-free motion and constraint satisfaction.
4. **Execution and receding horizon:** The safe control action is applied to the robot, and the planning horizon is shifted forward for the next iteration.

2.3 Key Features

- **Policy guidance:** The learned policy provides informed initialization, reducing the number of samples needed for good performance.
- **Safety guarantees:** The safety filter ensures that all applied control actions respect safety constraints, such as collision avoidance.
- **Modular design:** The MPPI update is encapsulated as a function, making the algorithm structure cleaner and more modular.
- **Receding horizon:** The algorithm continuously replans as new state information becomes available, enabling adaptation to disturbances and model uncertainties.

3 MPPIUpdate Function

The MPPIUPDATE function encapsulates the standard MPPI sampling and update process:

Algorithm 2 MPPIUPDATE Function

Require: Current state x_t , nominal sequence U , dynamics f , costs ℓ, ℓ_f , horizon H , samples K , temperature λ , covariance Σ , constraints $\mathcal{U}$

Ensure: Updated control sequence U

```
1: for  $k = 1$  to  $K$  do
2:    $x_{k,0} \leftarrow x_t$ 
3:    $S_k \leftarrow 0$ 
4:   for  $i = 0$  to  $H - 1$  do
5:     Sample noise  $\epsilon_{k,i} \sim \mathcal{N}(0, \Sigma)$ 
6:      $u_{k,i} \leftarrow \Pi_{\mathcal{U}}(u_i + \epsilon_{k,i})$ 
7:      $x_{k,i+1} \leftarrow f(x_{k,i}, u_{k,i})$ 
8:      $S_k \leftarrow S_k + \ell(x_{k,i}, u_{k,i})$ 
9:   end for
10:   $S_k \leftarrow S_k + \ell_f(x_{k,H})$ 
11: end for
12:  $\beta \leftarrow \min_k S_k$ 
13: for  $k = 1$  to  $K$  do
14:   $\tilde{w}_k \leftarrow \exp(-\frac{1}{\lambda}(S_k - \beta))$ 
15: end for
16:  $w_k \leftarrow \tilde{w}_k / \sum_{j=1}^K \tilde{w}_j$ 
17: for  $i = 0$  to  $H - 1$  do
18:   $u_i \leftarrow u_i + \sum_{k=1}^K w_k \epsilon_{k,i}$ 
19:   $u_i \leftarrow \Pi_{\mathcal{U}}(u_i)$ 
20: end for
21: return  $U$ 
```

4 Applications

Simplified PG-MPPI is particularly suitable for:

- Robotic manipulator planning in cluttered environments
- Tasks requiring both goal-directed behavior and safety guarantees
- Situations where learned policies can provide good initial guesses but may not be fully reliable
- Real-time applications where computational efficiency is crucial
- Systems where modular algorithm design is preferred for implementation and debugging